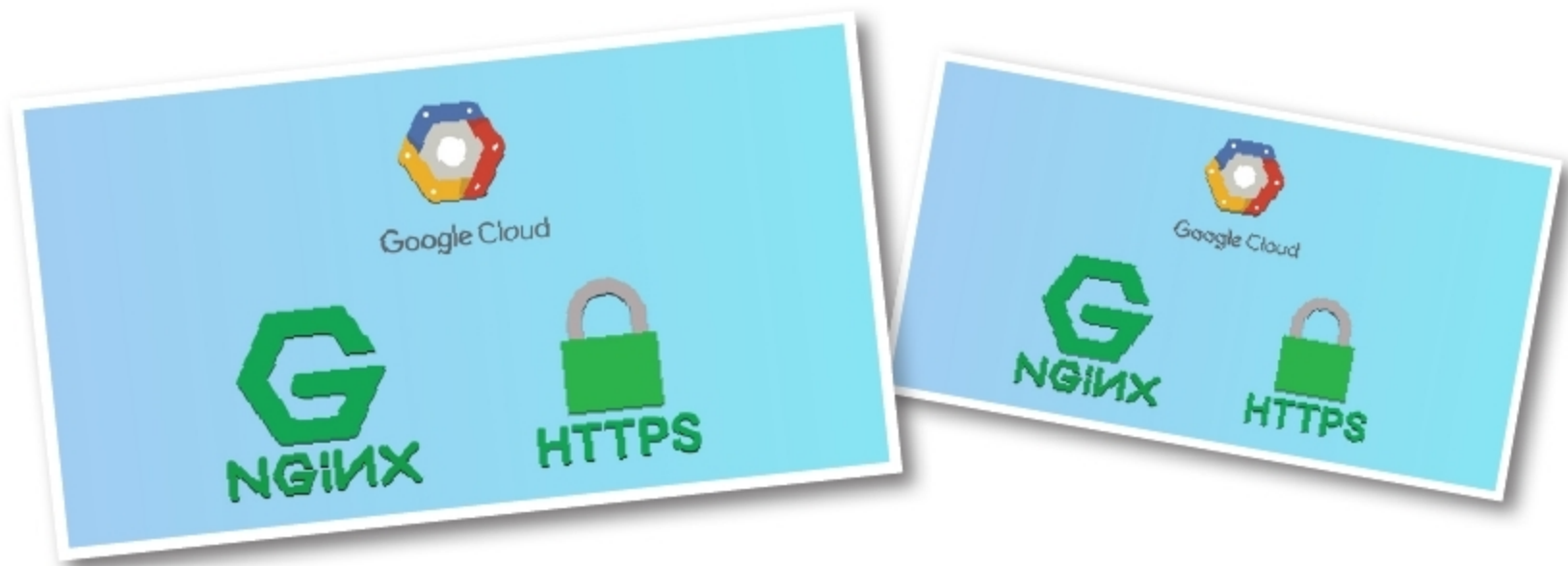


Deploy Django with SSL and Nginx on Google Cloud Platform Using Ubuntu VM



This article is a sequel to the earlier one on deploying Django 'on GCP, which was carried in the December 2019 issue of *OSFY*. It guides users on installing Nginx, enabling SSL and controlling the Django application using Supervisor to ease the process.

As you know, Nginx (<http://nginx.org/>) is an HTTP server which listens to Port 80 and can redirect to your application. We can also configure the SSL certificate and related configurations in Nginx.

```
sudo apt-get install nginx
```

You can check the status of Nginx by running the following command:

```
sudo systemctl status nginx.service
```

Creating an SSL certificate

We have different authorities providing SSL certificates—GoDaddy, Comodo, DigiCert, etc, but here, we are going to use Let's Encrypt (<https://letsencrypt.org/>)—an organisation

that provides free SSL certificates. (Let's Encrypt is a free, automated, and open certificate authority brought to you by the non-profit Internet Security Research Group.)

I have used ZeroSSL (<https://zerossl.com/>) to access Let's Encrypt, but have made the process simple to create, verify and download from the ZeroSSL online tool. Once the certificate is generated, you may *scp* the certificate to your GCP VM, where you have your Django application running.

Making changes in Django settings

For a production environment, you have to change *debug= true* to false, as follows:

```
debug=false
```

You may also add the security parameters given below to your Django settings:

```
SECURE_PROXY_SSL_HEADER = ('HTTP_X_FORWARDED_PROTO', 'https')
SECURE_SSL_REDIRECT = True
SESSION_COOKIE_SECURE = True
CSRF_COOKIE_SECURE = True
```

Installing and configuring Supervisor

Supervisor helps us to automate the starting and stopping of the Django process in production, as follows:

```
sudo apt-get install supervisor
```

Configure the Django app for Supervisor in the */etc/supervisor/config.d/<<your project name>>* file